

Juan David Munoz Bolanos

Junior Optical System Engineer

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Summary

Junior system engineer with hands-on lab work building, aligning, and calibrating optical laser systems using spatial light modulators and holography. I'm comfortable with PC and control technology, used to digging into technical problems systematically.

Research/Professional Experience

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Optical Systems Engineer / PhD Researcher

Optomechatronic System Design: Designed and built a multiphoton microscope and an automated wavefront sensing system from scratch, integrating lasers, spatial light modulators, and deformable mirrors.

Data Analysis & Algorithms: Developed Python and C++ pipelines for real time signal processing, iterative phase retrieval, and wavefield simulations to optimize system performance.

Systems Engineering & Quality: Applied V Model methodologies, Design of Experiments, and FMEA to track requirements and systematically resolve technical defects during prototype iterations.

Lab Management & Safety: Managed a high power laser laboratory, ensuring EH&S compliance, maintaining equipment logs, and coordinating component sourcing with suppliers like Thorlabs and Edmund Optics.

Aug 2018 to Oct 2019 **INTECOL S.A.S.** *Medellín, Colombia*
Junior Machine Vision Engineer (Field Service)

Machine Vision Pipelines: Engineered and deployed Halcon and Cognex VisionPro image processing pipelines for high speed industrial production lines, improving quality control.

Illumination Design: Designed custom bright field and dark field illumination systems to reliably detect defects, liquid levels, and sealing states on fast moving products.

Industrial Automation: Integrated vision systems with robotic arms and PLC controllers, executing on site testing, validation, and customer sign off.

Technical Competencies

Software & Tools

PythonC++LabVIEWHalconOpenCVZEMAXMATLABGit

Optical Systems

Adaptive OpticsWavefront SensingMicroscopySpectroscopyLaser SystemsDMDs

Systems & Quality

V-Model8D Problem SolvingDesign of ExperimentsISO 9001

Education

Jan 2022 to Jun 2026 **Medical University of Innsbruck** *Innsbruck, Austria*
Doctor of Philosophy in Optical Metrology

Oct 2019 to Sep 2021	Friedrich Schiller Univ. Jena Master of Science in Photonics	Jena, Germany
Aug 2012 to Aug 2018	University of Cauca Bachelor of Science in Physics Engineering	Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

- Biking/Hiking
- Bachata Dancing
- Harmonica
- Learning German

Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte Medical University of Innsbruck monika.ritsch-marte@i-med.ac.at	PD Dr. Christoph Krafft Leibniz IPHT christoph.krafft@leibniz-ipht.de
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